Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following is not permissible arrangement of electrons in an atom?

(A) $n = 5, l = 3, m = 0, s = +1/2$

(B) $n = 3, l = 2, m = -3, s = -1/2$

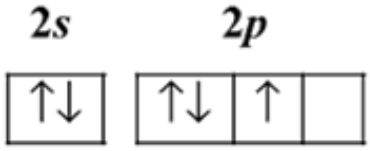
(C) $n = 3, l = 2, m = -2, s = -1/2$

(D) $n = 4, l = 0, m = 0, s = -1/2$

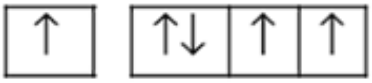
Q.2

The orbital diagram in which the Aufbau principle is violated is:

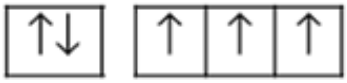
(A)



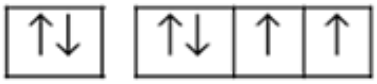
(B)



(C)



(D)



Q.3

Among the following groupings which represents the collection of isoelectronic species?

(A) $NO^+, C_2^{2-}, O_2^-, CO$

(B) N_2, C_2^{2-}, CO, NO

(C) CO, NO^+, CN^-, C_2^{2-}

(D) NO, CN^-, N_2, O_2^-

Q.4

Let m_p be the mass of a proton, m_n that of a neutron, M_1 that of a $^{20}_{10}Ne$ nucleus and M_2 that of a $^{40}_{20}Ca$ nucleus. Then

(A) $M_2 = 2M_1$

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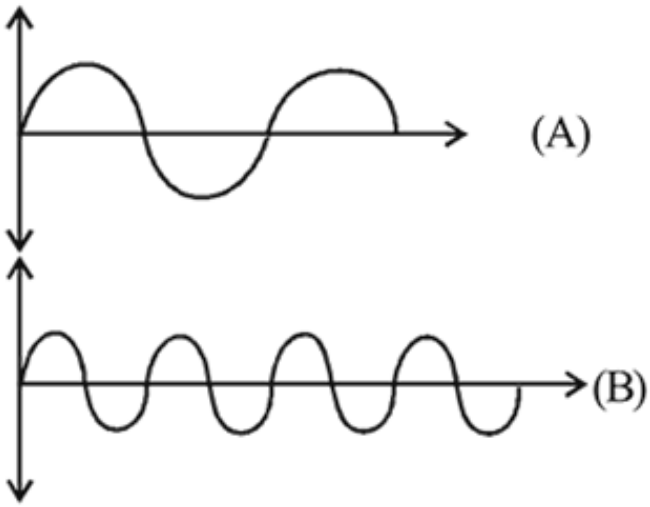
(B) $M_1 < 10\,(\,m_p + m_n)$

(C) $M_2 > 2M_1$

(D) $M_1 = M_2$

Q.5

What will be the difference between electromagnetic radiation shown in A and B respectively ?



(i) Velocity
(ii) Wavelength
(iii) Frequency
(iv) Energy

(A) (ii) only

(B) (ii) and (iv)

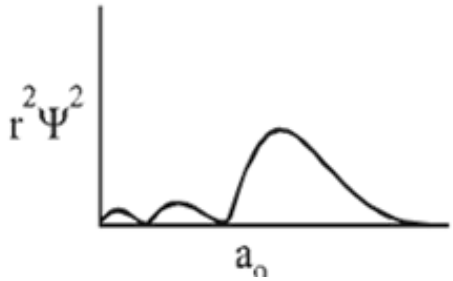
(C) (ii), (iii) and (iv)

(D) (iv) only

Q.6

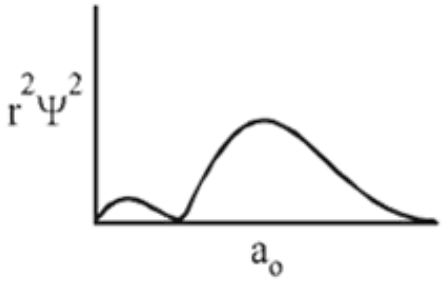
Which of the following radial distribution graphs correspond to $l = 2$ for the H atom ?

(A)

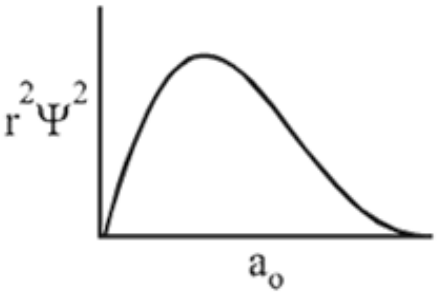


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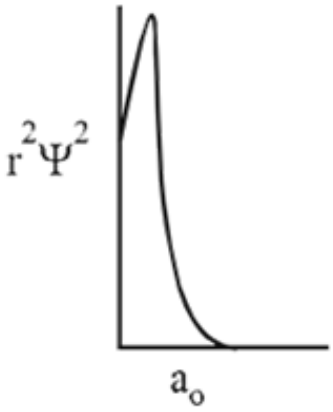
(B)



(C)



(D)



Q.7

Which combinations of quantum numbers, $\mathrm{n}, l, \mathrm{m}$ and s for the electron in an atom does not provide a permissible solution of the wave equation?

(A) $3, 2, 1, +\frac{1}{2}$

(B) $3, 1, 1, -\frac{1}{2}$

(C) $3, 3, 1, -\frac{1}{2}$

(D) $3, 2, -2, +\frac{1}{2}$

Q.8

In a multi-electron atom, which of the following orbitals described by the three quantum members will have the same energy in the absence of magnetic and electric fields?

(A) $n = 1, l = 0, m = 0$

(B) $n = 2, l = 0, m = 0$

(C) $n = 2, l = 1, m = 1$

(D) $n = 3, l = 2, m = 1$

(E) $n = 3, l = 2, m = 0$

(A) (D) and (E)

(B) (C) and (D)

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(C) (B) and (C)

(D) (A) and (B)

Q.9

Which of the following represents Schrödinger equation/s

(A) $\left(\frac{\partial^2}{\partial \mathrm{x}^2}+\frac{\partial^2}{\partial \mathrm{y}^2}+\frac{\partial^2}{\partial \mathrm{z}^2}\right) \Psi+\frac{8 \pi^2 \mathrm{m}}{\mathrm{h}^2}[\mathrm{E}-\mathrm{V}] \Psi=0$

(B) $\hat{\mathrm{H}} \Psi=\mathrm{E} \Psi$

(C) $(\hat{\mathrm{T}}+\hat{\mathrm{V}}) \Psi=\mathrm{E} \Psi$

(D) All of them

Q.10

Which of the following should be the number of electrons present in X^{2+} on the basis of electronic configuration, if the ion X^{3-} has 14 protons?

(A) 12

(B) 14

(C) 16

(D) 18

Q.11

Which one of the following set of quantum numbers is not possible for $4p$ electron?

(A) $\mathrm{n}=4, l=1, \mathrm{~m}=-1, \mathrm{~m_S}=+\frac{1}{2}$

(B) $\mathrm{n}=4, l=1, \mathrm{~m}=0, \mathrm{~m_S}=+\frac{1}{2}$

(C) $\mathrm{n}=4, l=1, \mathrm{~m}=2, \mathrm{~m_S}=+\frac{1}{2}$

(D) $\mathrm{n}=4, l=1, \mathrm{~m}=-1, \mathrm{~m_S}=-\frac{1}{2}$

Q.12

If the nitrogen atom had electronic configuration $1 \mathrm{~s}^7$ it would have energy lower than that of the normal ground state configuration $1 \mathrm{~s}^2 2 \mathrm{~s}^2 2 \mathrm{p}^3$ because the electrons would be closer to the nucleus. Yet $1 \mathrm{~s}^7$ is not observed. It violates

(A) Heisenberg's uncertainty principle

(B) Hund's rule

(C) Pauli exclusion principle

(D) Bohr postulate of stationary orbits

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Q.13

The correct order of increasing energy of atomic orbitals is

(A) $5\mathrm{p} < 4\mathrm{f} < 6\ \mathrm{s} < 5\ \mathrm{d}$

(B) $5\mathrm{p} < 6\ \mathrm{s} < 4\mathrm{f} < 5\ \mathrm{d}$

(C) $4\mathrm{f} < 5\mathrm{p} < 5\ \mathrm{d} < 6\ \mathrm{s}$

(D) $5p < 5d < 4f < 6s$

Q.14

The ratio of magnetic moments of $\mathrm{Fe(III)}$ and $\mathrm{Co(II)s}$

(A) $7 : 3$

(B) $3 : 7$

(C) $\sqrt{7} : \sqrt{3}$

(D) $\sqrt{3} : \sqrt{7}$

Q.15

The correct set of four quantum numbers for the valence electrons of rubidium atom $(Z = 37)$ is:

(A) $5, 0, 0, +\frac{1}{2}$

(B) $5, 1, 0, +\frac{1}{2}$

(C) $5, 1, 1, +\frac{1}{2}$

(D) $5, 0, 1, +\frac{1}{2}$

Q.16

Charge to mass ratio of cathode rays depends on

(A) Nature of the gas in the discharge tube

(B) Pressure of the gas in the discharge tube

(C) Material of cathode

(D) None of these

Q.17

Which of the following symbols correctly represents an orbitals in an atom?

(A) $\Psi_{n,l,m} = R_n\Theta_l\Phi_m$

(B) $\Psi_{n,l,m} = R_{n,l}\Theta_l\Phi_m$

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(C) $\Psi_{n,l,m} = R_{n,l}\Theta_{l,m}\Phi_m$

(D) $\Psi_{n,l,m} = R_{n,l}\Theta_{l,|m|}\Phi_m$

Q.18

Which of the following statements on the square of atomic wave function at a point near to the nucleus is not correct?

(A) Ψ^2 may be positive, negative or imaginary

(B) Ψ^2 is proportional to electron density

(C) Ψ^2 is directly proportional to the probability of finding the electron

(D) Ψ^2 is equal to the probability of finding the electron if Ψ is a normalized wave function

Q.19

The wave function of atomic orbital of H like species is given as

$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} Z^{\frac{3}{2}} 2 - Zre^{-\frac{Zr}{2}}$

The radius for nodal surface for He^+ ion in Å .

(A) 1

(B) 2

(C) 3

(D) 4

Q.20

When 4 f subshell is completely filled with electrons, the next electron will enter into a subshell for which, $n + L$ value is equal to

(A) 7

(B) 5

(C) 6

(D) 8

Answers & Solutions

Q.1 Answer:

$n = 3, l = 2, m = -3, s = -1/2$

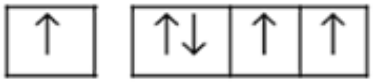
Solution:

$m = -l$ **to** $+l$**, through zero thus for** $l = 2$ **values of** m **will be** $-2, -1, 0, +1, +2$

Therefore for $l = 2, m$ **cannot have the value** -3

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Q.2 Answer:



Solution:

According to Aufbau principle, the orbital of lower energy $(2s)$ should be fully filled before the filling of orbital of higher energy starts.

Q.3 Answer:

$\mathrm{CO}, \mathrm{NO}^{+}, \mathrm{CN}^{-}, \mathrm{C}_2^{2-}$

Solution:

The species $\mathrm{CO}, \mathrm{NO}^{+}, \mathrm{CN}^{-}$ and C_2^{2-} contain 14 electrons each.

Q.4 Answer:

$\mathrm{M}_2 = 2\mathrm{M}_1$

Solution:

$^{20}_{10}$ Ne contains 10 protons and 10 neutrons

$\therefore \mathrm{M}_1 = 10\ \mathrm{m_p} + 10\ \mathrm{m_n}$

$^{40}_{20}$Ca contains 20 protons and 20 neutrons

$\therefore \mathrm{M}_2 = 20\ \mathrm{m_p} + 20\ \mathrm{m_n}$

$\therefore \mathrm{M}_2 = 2\mathrm{M}$

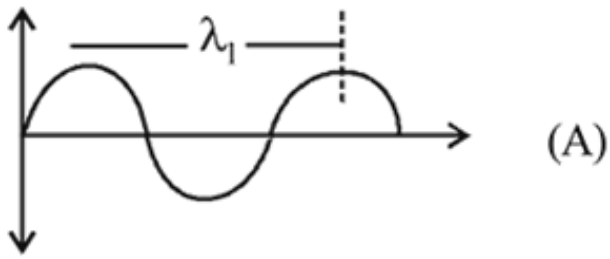
Q.5 Answer:

(ii), (iii) and (iv)

Solution:

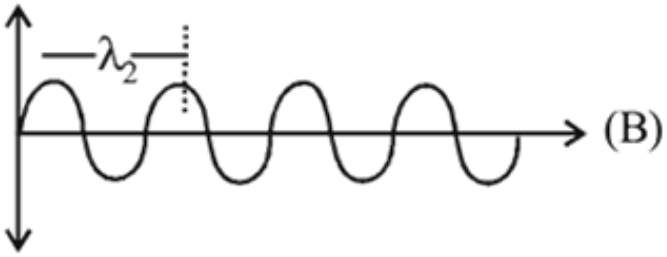
e/m waves shown in figure A has higher wavelength in comparison to e/m waves shown in figure B.

Thus these waves also differ in frequency and energy. $v = \frac{c}{\lambda}$



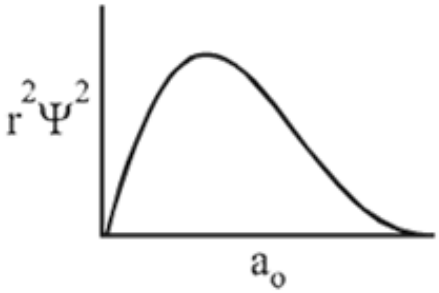
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$\Rightarrow E_1 = \frac{hc}{\lambda_1}$



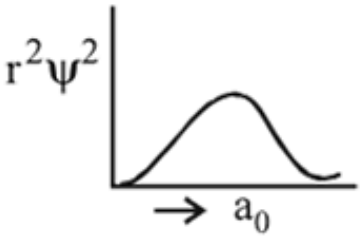
$\Rightarrow E_2 = \frac{hc}{\lambda_2} \quad \lambda_1 > \lambda_2 \Rightarrow E_1 < E_2$

Q.6 Answer:



Solution:

$l = 2$ **represent** d **orbital for which**



Q.7 Answer:

$3, 3, 1, -\frac{1}{2}$

Solution:

Possible values of l **and** m **depend upon the value of** n

$l = 0$ **to** $(n-1)$

$\mathrm{m} = -l$ **to** $+l$ **through zero**

$\mathrm{s} = +\frac{1}{2}$ **and** $-\frac{1}{2}$

Thus for $\mathrm{n} = 3$

l **may be 0,1 or** 2**; but not 3**

m may be $-2, -1, 0, +1$ **or** $+2$

s may be $+\frac{1}{2}$ **or** $-\frac{1}{2}$

Q.8 Answer:

(D) and (E)

Solution:

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The energy of an orbital is given by $(n+l)$. In (D) and (E), $(n+l)$ value is $(3+2)=5$ hence they will have same energy, since here n values are also same.

Q.9 Answer:

All of them

Solution:

No solution available

Q.10 Answer:

12

Solution:

X^{3-} has 14 protons, i.e.,
X also has 14 protons and therefore 14 electrons.
$\mathrm{X}=14=1s^2,2s^22p^6,3s^23p^2$
$\mathrm{X}^{2+}=1s^2,2s^22p^6,3s^2=12$ **electrons**

Q.11 Answer:

$\mathrm{n}=4, l=1,\ \mathrm{m}=2,\ \mathrm{m_S}=+\frac{1}{2}$

Solution:

For $4p$ electron $\mathrm{n}=4,1=1,\ \mathrm{m}=-1,0+1$ **and** $\mathrm{S}=+1/2$ **or** $-1/2$

Q.12 Answer:

Pauli exclusion principle

Solution:

Not more than two electrons can be present in same atomic orbital. This is Pauli's exclusion principle.

Q.13 Answer:

$5\mathrm{p}<6\ \mathrm{s}<4\mathrm{f}<5\ \mathrm{d}$

Solution:

According to Aufbau principle electron are filled in various atomic orbitals in the increasing order of energy: $1\ \mathrm{s}<2\ \mathrm{s}<2\mathrm{p}<3\ \mathrm{s}<3\mathrm{p}<4\ \mathrm{s}<3\ \mathrm{d}<4\mathrm{p}<5\ \mathrm{s}<4\ \mathrm{d}<5\mathrm{p}<6\ \mathrm{s}<4\mathrm{f}<5\ \mathrm{d}<6\mathrm{p}<7\ \mathrm{s}$

Q.14 Answer:

$\sqrt{7}:\sqrt{3}$

Solution:

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$\mathrm{Fe(III)} = [\mathrm{Ar}]3\ \mathrm{d}^5$ **unpaired electrons** $= 5$; **Magnetic moment** $= \sqrt{5(5+2)}$; **Ratio** $= \sqrt{7} : \sqrt{3}$

$\mathrm{Co(II)} = [\mathrm{Ar}]3\ \mathrm{d}^7$ **unpaired electrons** $= 3$

Magnetic moment $= \sqrt{3(3+2)}$

Ratio $= \sqrt{7} : \sqrt{3}$

Q.15 Answer:

$5, 0, 0, +\frac{1}{2}$

Solution:

The electronic configuration of Rubidium $(\mathrm{Rb} = 37)$ **is** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$

Since last electron enters in $5s$ **orbital**

Hence $n = 5, l = 0, m = 0, s = \pm\frac{1}{2}$

Q.16 Answer:

None of these

Solution:

Charge to mass ratio of cathode rays is independent of the nature of the gas or material of n

Q.17 Answer:

$\Psi_{n,l,m} = R_{n,l}\Theta_{l,|m|}\Phi_m$

Solution:

The function R **depends on the quantum numbers** n **and** l**, the function** Θ **depends on** l **and** $|m|$ **and the function** Φ **depends on** m**.**

Q.18 Answer:

Ψ^2 may be positive, negative or imaginary

Solution:

Ψ^2 **cannot be imaginary.** Ψ^2 **is positive for a real wave function and** $\Psi^*\Psi$ **(where** Ψ^* **is the complex conjugate of** Ψ**) is positive.**

Q.19 Answer:

1

Solution:

As at nodal surface probability of finding an electron is zero, therefor, $\psi = 0$

So $2 - Zr = 0$

$r = \frac{2}{Z} = \frac{2}{2} = 1$ Å

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Q.20 Answer:

7

Solution:

The next electron will enter into 5d subshell whose $(n + l)$ value is equal to 7.

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